Mandhagini Madhavan

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PROFESSIONAL SUMMARY

MSc Artificial Intelligence graduate with hands-on experience developing **AI, Machine Learning and Generative AI applications** using Python. Experienced in building **LLM-powered applications, RAG pipelines, multi-agent AI systems and REST APIs**, with practical experience across NLP, computer vision and data analysis. Skilled in **FastAPI, Streamlit, SQL, Git and Docker**, with experience developing end-to-end AI applications from data processing and model integration to API development and deployment. Seeking **Junior AI Engineer, Machine Learning Engineer or Graduate AI Engineer** opportunities.

TECHNICAL SKILLS

Programming: Python (pandas, NumPy), SQL, JavaScript, TypeScript
AI & Machine Learning: Machine Learning, Deep Learning, NLP, Computer Vision, Large Language Models (LLMs), RAG, Prompt Engineering, Semantic Search, Multi-Agent AI Systems
Frameworks & Libraries: TensorFlow, PyTorch, scikit-learn, OpenCV, FastAPI, Streamlit, React
AI & Data Engineering: REST API Development, Model Integration, Data Preprocessing, Feature Engineering, Vector Embeddings, API Integration
Development & Tools: Git, GitHub, Docker, pytest, FFmpeg, Jupyter Notebook, VS Code
Data & Visualisation: pandas, NumPy, Matplotlib, Seaborn, Excel (Pivot Tables, VLOOKUP)
Cloud: AWS (S3, EC2 — basic exposure)

PROJECT EXPERIENCE

InsightForge Multi-Agent AI Investigation Platform
- Developing an AI investigation platform using **Python, FastAPI, React, RAG and LLM-based multi-agent workflows**.
- Built REST APIs, evidence retrieval and semantic search workflows with **Docker-based deployment**.

TimeSync AI — Intelligent Timestamp Reconciliation & Video Clipping
- Developed a Python system that reconciles conflicting **video and STT timestamps** using confidence-based decision logic.
- Integrated **FFmpeg**, silence-aware clipping, automated testing with **pytest**, CLI/UI support and Docker deployment.

Wildfire Detection System — Computer Vision & NLP
- Developed a multimodal wildfire detection system using **CNN, MobileNetV2, ResNet50 and Attention CNN** for image classification.
- Combined image predictions with **NLP sentiment analysis** and automated alert generation for early wildfire detection.

Multi-Persona AI Assistant
- Built an **LLM-powered Streamlit application** that generates context-aware responses across multiple professional personas.
- Implemented **prompt engineering, conversation memory and API integration** for real-time personalised responses.

PROFESSIONAL EXPERIENCE

Clerical Marker, Cambridge University Press & Assessment 05/2024 – 08/2024 | United Kingdom
- Reviewed and validated high volumes of assessment data while maintaining accuracy and consistency against defined quality standards.
- Applied strong attention to detail and data-validation practices when handling structured assessment information.

AI & Data Science Intern, Freelance 01/2023 – 08/2023 | India
- Developed Python-based machine learning and data-analysis projects involving data preprocessing, model development and performance evaluation.
- Worked with real-world datasets to perform data cleaning, exploratory analysis and model optimisation.

EDUCATION

Bachelor of Computer Applications (BCA), Bishop Heber College India (Completed: 2023)

MSc Artificial Intelligence, Anglia Ruskin University United Kingdom (Completed: 2025)

CERTIFICATIONS

AWS Academy Machine Learning Foundations

LANGUAGES

Tamil — Native/Bilingual **English** — Fluent